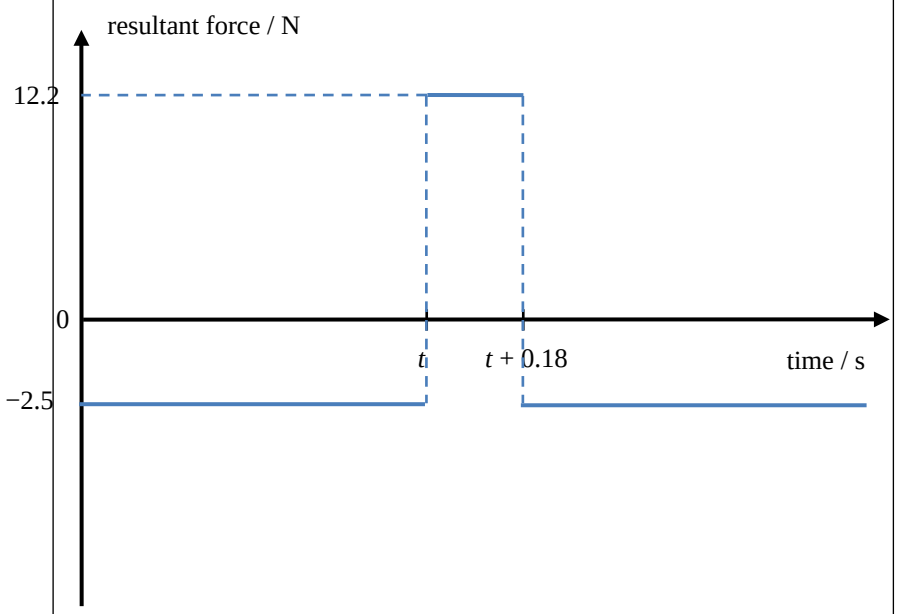
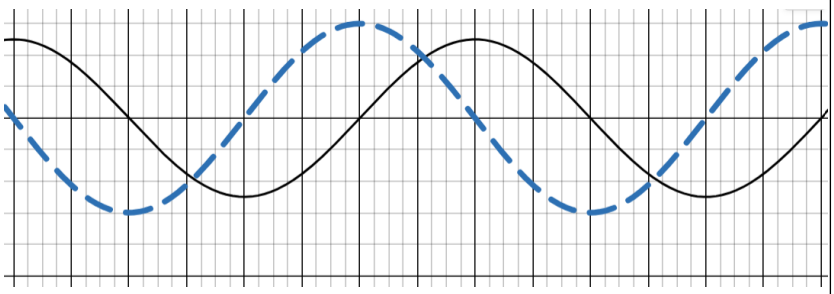
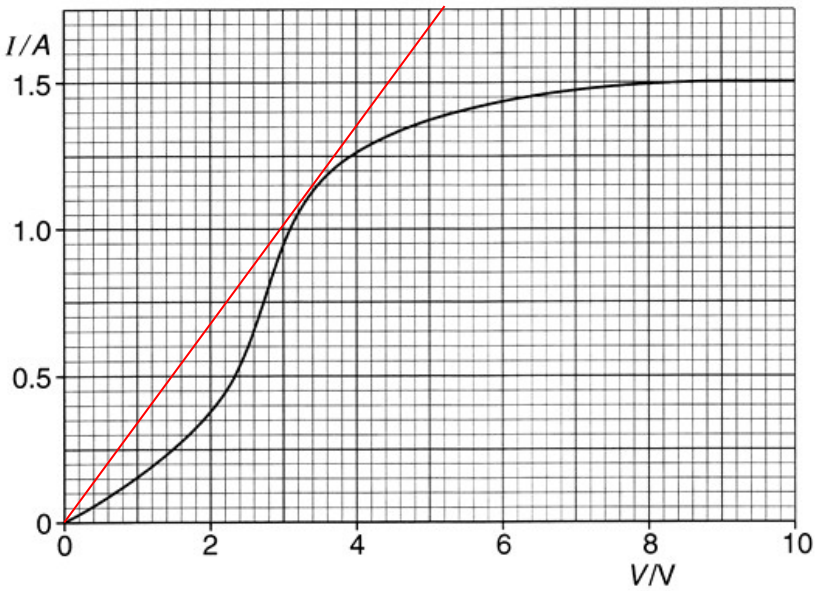
Answers to 2026 JC2 Preliminary Examination Paper 3 (H2 Physics)

Suggested Solutions:

No.	Solution	Remarks
1(a)	Object A: $s = ut + \frac{1}{2}at^2$ $-14 = 0 + \frac{1}{2}(-9.81)t^2$ $t = 1.6894$ s Object B: $s = ut + \frac{1}{2}at^2$ $-3.6 = u(1.6894) + \frac{1}{2}(-9.81)(1.6894)^2$ $u = 6.1560 \approx 6.2$ m s^{-1}	[1] for t [1] for correct equation and substitution to get u [1] for correct answer
1(b)(i)	The initial vertical component of velocity for B is smaller, it will reach lower height. It will take a shorter time.	[1] for correct explanation [1] for shorter time
1(b)(ii)	Since height of release and initial speed remain unchanged, the loss in gravitational potential energy is the same, regardless of angle. By conservation of energy, the gain in kinetic energy is the same and hence final speed will remain unchanged.	[1] for any correct explanation [1] for same time
2(a)	The (total) kinetic energy changes / decreases. Therefore the collision is inelastic. OR The relative speed of approach is not equal to / greater than the relative speed of separation. Therefore the collision is inelastic.	[1]
2(b)(i)	$\Delta p = (0.25)\left[3.6 - (-5.2)\right]$ $= 2.2$ kg m s^{-1} Change in momentum of the ball takes place vertically upwards.	[1] for correct answer [1] for correct direction

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2(b)(ii)1.	$F_{\text{by ground on ball}} - mg = \frac{\Delta p}{\Delta t}$ $F_{\text{by ground on ball}} - (0.25)(9.81) = \frac{2.2}{0.18}$ $F_{\text{by ground on ball}} = 14.6747 \text{ N} = 15 \text{ N}$ $F_{\text{by ball on ground}} = 15 \text{ N}$	[1] for correct Newton's second law equation [1] for correct substitution [1] for Newton's third law and for correct answer
2(b)(ii)2.		[1] for two horizontal straight lines: one before *t* and one after *t* + 0.18 s; same sign and same magnitude [1] for one horizontal line between *t* and *t* + 0.18 s; opposite sign compared with the two previous lines [1] for correct numerical values labelled on the resultant force axis
3(a)	Moment of weight about A $= 65 \times (0.30 \times \cos 30^o)$ $= 16.88$ $\approx 16.9 \text{ N}$ Taking moment about A, Clockwise moment = Anti-clockwise moment $65 \times (0.30 \times \cos 30^o) = F \sin 50^o \times 0.60$ $\Rightarrow F = 36.74 \approx 36.7 \text{ N}$	[1] ans [1] sub [1] ans
3(b)	When the angle decreases, <u>the perpendicular distance from A to the line of action of force *F* decreases</u>. The <u>magnitude of *F* will then increase</u> to <u>balance</u> the same <u>clockwise moment by the weight of the bar.</u>	[1] expl [1] state

No.	Solution	Remarks
3(c)	Horizontally, $R_x = F\cos 20^o = 36.74 \times \cos 20^o = 34.53 \approx 34.5$ N Vertically, $R_y + F\sin 20^o = 65$ $\Rightarrow R_y = 52.43 \approx 52.4$ N $R = \sqrt{34.53^2 + 52.43^2} = 62.78$ ≈ 62.8 N	[1] value for R_x [1] value for R_y [1] ans
4(a)	Polarisation is the process of restricting a wave's vibrations to a single **<u>plane perpendicular to the direction the wave</u>**. Longitudinal waves cannot be polarised because their oscillation occurs **<u>parallel to the direction of propagation</u>**, leaving no perpendicular component to orient or filter.	[1] Explain polarisation [1] Correct explanation
4(b)(i)		[1] Sinusoidal graph of similar frequency [1] Phase lead of 90 or -sin shape
4(b)(ii)	$\phi = \frac{x}{\lambda}(2\pi)$ $3.5 = \frac{0.300}{\lambda}(2\pi)$ $\lambda = 0.5383$ $v = f\lambda$ $v = (650)(0.5383)$ $v = 350$ m s^{-1}	[1] for correct calculation of λ [1] for correct calculation of v
4(b)(iii)	The loudspeaker is **<u>not a point source</u>**. The amplitude decreasing inversely with distance only applies for a point source.	[1] correct explanation
5(a)(i)	Current flowing through the circuit $I = \frac{4.0}{48} = 0.0833$ A Using $E = V + Ir$ $4.5 = 4.0 + (0.0833)(r)$ $r = 6.0\ \Omega$	[1] value for I [1] ans

No.	Solution	Remarks
5(a)(ii)	Energy dissipated $=\left(\frac{V^2}{R}\right)t$ $=\left(\frac{4.0^2}{48}\right)(2\times60)$ $=40$ J	[1] ans
5(b)(i)	 Minimum resistance $=\frac{3.4}{1.15}=2.96$ $\approx 3.0\ \Omega$	[1] line drawn [1] working to calculate resistance
5(b)(ii)	Effective resistance of 48 Ω resistor and electrical component C connected in parallel will decrease to a value lower than 48 Ω. Hence, the potential diffeence across the 48 Ω resistor will drop, resulting in a decrease in the energy dissipated.	[1] explanation
5(c)(i)	Gradient $=\frac{3.6-1.9}{0.8-0.4}=4.25$	[1] ans

No.	Solution	Remarks
5(c)(ii)	$R = \frac{\rho L}{A}$ $\frac{\rho}{A}$ represents the gradient of graph $\frac{\rho}{A} = 4.25$ $\rho = 4.25 \times 0.12 \times 10^{-6}$ $= 5.1 \times 10^{-7}\ \Omega$ m	[1] ans
6(a)	Magnetic flux linkage $\Phi = NB_{\perp}A$ $= (250)(3.0 \times 10^{-2})(2.4 \times 10^{-4})$ $= 1.8 \times 10^{-3}$ Wb (shown)	[1] for correct numerical substitution
6(b)(i)	One complete revolution of the coil takes 0.50 s Magnitude of induced e.m.f. $E = \left\lvert \frac{\Delta\Phi}{\Delta t} \right\rvert$ $= \left\lvert \frac{\Phi_f - \Phi_i}{\Delta t} \right\rvert$ $= \left\lvert \frac{(250)(3.0 \times 10^{-2} \cos 45)(2.4 \times 10^{-4}) - 1.8 \times 10^{-3}}{\frac{1}{8} \times 0.50} \right\rvert$ $= 8.435 \times 10^{-3}$ $\approx 8.44 \times 10^{-3}$ V	[1] for correct numerical substitution [1] for correct answer
6(b)(ii)	As the coil rotates, the <u>component of the magnetic flux density perpendicular to the plane of the coil varies with the cosine of the angle θ</u>. This implies that the magnetic flux linkage continuously fluctuates between a maximum value and zero following a sinusoidal pattern. According to Faraday's Law, the induced e.m.f. is determined by the rate of change this perpendicular component of the magnetic flux. Hence, the <u>rate of change magnetic flux linkage through the coil results in a sinusoidal pattern</u>, resulting in a sinusoidal induced e.m.f..	[1] [1]
6(c)(i)	Lenz's law states that the <u>polarity/direction</u> of the induced e.m.f. is such that it tends to produce a current with effects that oppose the <u>change</u> in flux that caused the induced e.m.f.	[1]
6(c)(ii)	As solenoid P is withdrawn, the <u>magnetic flux linkage in P decreases</u>, inducing an e.m.f. by Faraday's law and causing a	[1]

No.	Solution	Remarks
	current to flow through P since its terminals form a closed circuit. By Lenz's law, this induced current creates its own magnetic field that opposes the motion causing it. The interaction between the magnetic field of P and the magnetic field of Q creates an attractive magnetic force that opposes the withdrawal, so work must be done (a force is needed) to pull P out.	[1] [1]
7(a)	$\|\psi\|^2$ is the probability density for finding the particle at position x.	[1] Correct explanation
7(b)(i)	Boundary reflections cause opposite travelling matter-waves to superpose and form standing waves. Infinite potential at boundary cause wave to be zero amplitude.	[1] formation of standing waves. [1] Boundary amplitude
7(b)(ii)	Square the wavefunction for $n = 3$  Probability is 1/6 or 0.167, as area from L/3 to L/2 is 1/6.	[1] Any correct method / sound reasoning [1] Correct answer
7(c)	$\Delta E = E_2 - E_1$ $\Delta E = (2^2 - 1^1)\frac{h^2}{8mL^2}$ $\Delta E = \frac{3h^2}{8mL^2}$ $\Delta E = \frac{3(6.63x10^{-34})^2}{8(9.11x10^{-31})(2x10^{-10})^2}$ $\Delta E = 4.52x10^{-18}$ J	[1] Correct expression for ΔE [1] Correct answer
8(a)(i)	A field of force is a region of space where a body possessing a particular property experiences a force due to the presence of other bodies with the same property, without any physical contact between them.	[1]
8(a)(ii)	Electric field strength at a point in an electric field is defined as the electric force per unit charge on a positive charge placed at that point.	[1]
8(a)(iii)	The test particle must be stationary so that it experiences only the electric force due to the electric field. If it were moving, it could experience an additional magnetic force, causing the measured force not to be solely due to the electric field.	[1] for mentioning magnetic force
8(b)(i)	The electric field strength at a point in a field is equal to the	[1]

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	negative potential gradient at that point. Note: Need to explain the terms if $E = -\frac{dV}{dx}$ is used.	
8(b)(ii)1.	The electric field at point P is in the positive x direction (i.e. from sphere A to sphere B). Electric field strength is in the direction of decreasing potential.	[1] for correct direction [1] for correct explanation
8(b)(ii)2.	At x = 35.0 cm, the potential gradient $\frac{dV}{dx} = 0$. Hence the electric field strength $E = -\frac{dV}{dx} = 0$. The electric force on an electron $F = eE = 0$ N.	[1] for gradient [1] for zero electric field strength [1] for zero force
8(b)(ii)3.	Inside the sphere ($x = 0$ to $x = 9.0$ cm), the electric field strength is zero ($E = 0$) for static charge in a conductor. Since the electric field strength is equal to the potential gradient ($E = -\frac{dV}{dx}$), a zero field strength implies the electric potential gradient is zero, so electric potential remains constant.	[1] zero electric field strength [1] for applying electric field strength being equal to potential gradient
8(c)	The minimum speed (and also kinetic energy) of the particle occurs at the point of minimum electrical potential, i.e. x = 35 cm and V = 295 V. Gain in KE = Loss in EPE $\frac{1}{2}mv^2 = 0 - qV$ $\frac{1}{2}(9.11\times10^{-31})v^2 = 0 - (-1.6\times10^{-19})(295)$ $v = 1.0180\times10^7 \approx 1.02\times10^7$ m s^{-1}	[1] for identifying the correct point for lowest potential [1] for correct equation/ substitution [1] for correct answer
8(d)(i)	Time constant $\tau = RC$ $= (45)(120\times10^{-6})$ $= 0.0054$ s Use $Q = Q_0\left(1 - e^{-\frac{t}{\tau}}\right)$	[1] for correct time constant [1] for correct

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	$\frac{Q}{Q_0} = 1 - e^{-\frac{t}{\tau}}$ $0.95 = 1 - e^{-\frac{t}{0.0054}}$ $-\left(\frac{t}{0.0054}\right) = \ln(0.05)$ $t = 0.01617 \approx 0.016$ s	substitution [1] for correct answer
8(d)(ii)	Use $Q = CV$, Charge $Q = (120 \times 10^{-6})(240)$ $= 0.0288$ C So charge delivered by capacitor $= 0.95 \times 0.0288$ $= 0.0274$ C Use $E = \frac{Q^2}{2C}$, Energy $= \frac{(0.0274)^2}{2(120 \times 10^{-6})}$ $= 3.128$ ≈ 3.2 J	[1] for correct total charge [1] for correct charge delivered by capacitor [1] for correct answer
9(a)(i)	$-(A - 4)v + 4V = 0$	[1]
9(a)(ii)	$\frac{\text{initial kinetic energy of } \alpha\text{-particle}}{\text{initial kinetic energy of daughter nucleus}}$ $= \frac{\frac{1}{2}(4)V^2}{\frac{1}{2}(A - 4)v^2}$ $= \frac{(4)V^2}{(A - 4)v^2}$ From (i), $\frac{V}{v} = \frac{A - 4}{4}$ so $\frac{V^2}{v^2} = \frac{(A - 4)^2}{4^2}$ Therefore $\frac{(4)V^2}{(A - 4)v^2} = \frac{4}{(A - 4)}\frac{(A - 4)^2}{4^2} = \frac{A - 4}{4} = \left(\frac{1}{4}A - 1\right)$	[1] for correct ratio [2] for all correct manipulations

No.	Solution	Remarks
9(b)(i)	Mass difference = (211.9459 – 207.9374 – 4.0015)u = 0.007u $E = (\Delta m)c^2 = (0.007 \times 1.66 \times 10^{-27})(3 \times 10^8)^2$ $= 1.0458 \times 10^{-12}$ J $= \frac{1.0458 \times 10^{-12}}{1.60 \times 10^{-19} \times 10^6}$ $= 6.5363$ ≈ 6.54 MeV	[1] for correct mass difference [1] for correct substitution [1] for final answer
9(b)(ii)	$\frac{\text{initial kinetic energy of } \alpha\text{-particle}}{\text{initial kinetic energy of daughter nucleus}}$ $= \frac{1}{4}A - 1 = \frac{1}{4}(212) - 1 = 52$ $E_\alpha = \frac{52}{53}(6.54) = 6.4166 \approx 6.42$ MeV	[1] for correct ratio [1] for correct method
9(c)(i)	Some of the energy is released in the form of gamma radiation. (do not accept heat/ KE)	[1]
9(c)(ii)	The gamma photon has momentum and it is not likely to move along the direction of the thallium nucleus or the alpha particle. Since (total) momentum must be conserved and the gamma photon moves away with some momentum in random direction, it is likely that the thallium nucleus and the alpha particle do not move off in opposite directions.	[1] for momentum of gamma photon [1] for random direction of this gamma photon [1] for conservation of momentum
9(d)(i)1.	The half-life $t_{\frac{1}{2}}$ of a radioactive nuclide is the time for the number of radioactive nuclide to reduce to half of its initial value.	[1]
9(d)(i)2.	The decay constant λ is the fraction of the total number of atoms that decay per unit time.	[1]
9(d)(ii)	Using $N = N_0 e^{-\lambda t}$, after time $t_{\frac{1}{2}}$, the number of undecayed atoms N would be $\frac{N_0}{2}$. Therefore, $\frac{N_0}{2} = N_0 e^{-\lambda t_{\frac{1}{2}}}$ $2 = e^{\lambda t_{\frac{1}{2}}}$ $\ln 2 = \lambda t_{\frac{1}{2}} \Rightarrow \lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$	[1] for using the correct equation and substitution [1] for correct method

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9(d)(iii)	Half-life of bismuth-212 $=\frac{\ln 2}{\lambda}$ $=\frac{\ln 2}{1.9\times10^{-4}}$ $=3648.1$ ≈ 3650 s	[1]
9(d)(iv)	1st observation: Using $\frac{x}{x_0}=\left(\frac{1}{2}\right)^{\frac{t}{t_{1/2}}}=\left(\frac{1}{2}\right)^{\frac{2(60)(60)}{3648.1}}=0.25461$ After two hours, the number of bismuth-212 nuclei is quartered, meaning that two half-lives had passed. 2nd observation: Bismuth-212 disintegrates to form thallium-208 which has half-life 190 s, or about 3 minutes. After two hours it is found that the number of thallium-208 nuclei is very much less than $\frac{3}{4}N$ because of its very short half-life. the newly formed Thallium-208 is not stable and disintegrates.	[1] for explaining why the number of bismuth nuclei would be reduced to approximately $\frac{1}{4}N$. [1] for explain short half-live, not stable and disintegrate